

Technical Scholarly Writing and Scientific Publication Skills

From Internship Reports to Journal Publications

Facilitator

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MODULE 1

INTRODUCTION TO SCIENTIFIC WRITING

Workshop Objectives

By the end of this workshop students should:

- ✓ Understand scientific writing
- ✓ Understand publications
- ✓ Identify quality journals
- ✓ Write publishable manuscripts
- ✓ Avoid predatory journals
- ✓ Respond to reviewers
- ✓ Convert projects into publications

Why
Network/AI
Engineers
Must Write

Engineers communicate:

- Designs
- Architectures
- Results
- Innovations
- Standards

Why Publish?

Benefits:

- Scholarships
- MSc admissions
- PhD admissions
- Career growth
- Visibility
- Citations
- Funding opportunities

Research Lifecycle

Problem Identification

Literature Review

Research Gap Identification

Research Objectives

Research Design

Data Collection

Data Analysis

Results & Discussion

Manuscript Writing

Journal Submission

Peer Review

Publication

Publication Pipeline

(Internship to Publication)

1. Internship Activities
2. Problem Observation
3. Final Year Project
4. Prototype Development
5. Experimental Evaluation
6. Conference Paper
7. Extended Journal Paper
8. MSc / PhD Research

Example

- ***Internship Activity:*** Installing WiFi networks.
- ***Research Question:*** How does throughput vary with distance?
- ***Publication:*** Performance Evaluation of WiFi Networks for Rural Connectivity.

MODULE 2

UNDERSTANDING SCIENTIFIC PUBLICATIONS

What is a Manuscript?

Definition: A manuscript is a structured scientific document prepared to communicate research findings for peer review and publication.

Purpose

A manuscript enables researchers to:

- Share knowledge
- Report discoveries
- Validate findings
- Establish credibility
- Enable reproducibility
- Contribute to science and engineering

Examples

- Conference paper
- Journal article
- Review paper
- Survey paper
- Technical note

Types of Publications

Type	Typical Pages	Purpose	Example
Conference Paper	4–8	Preliminary findings	LoRa Performance Evaluation
Journal Paper	8–25	Complete study	Adaptive Network Architecture
Review Paper	15–40	Summarize literature	Review of LoRa in Smart Agriculture
Survey Paper	15–30	Compare approaches	Survey of AI Models in Agriculture
Technical Note	2–6	Short innovation	New RSSI Measurement Method
Case Study	6–12	Real implementation	Smart Campus Network Deployment

Conference vs Journal Papers

Example

Conference: "LoRa
Network Evaluation for
Smart Agriculture"

Journal: "Comprehensive
Performance Evaluation
and Optimization of LoRa
Networks for Smart
Agriculture"

Feature	Conference	Journal
Length	Short	Long
Novelty	Moderate	High
Detail	Limited	Extensive
Review Time	Weeks	Months
Acceptance Rate	Higher	Lower
Citations	Moderate	Higher
Career Value	Good	Excellent

Examples of Types of Publications

- **Research Article:** Original contribution.

Example: Adaptive Network Selection for Smart Beehive Monitoring.

- **Review Article:** Summarizes existing work.

Example: Review of LoRa Communication Technologies for Agriculture.

- **Survey Paper:** Compares approaches.

Example: Survey of Machine Learning Models for Crop Prediction.

- **Case Study:** Real-world implementation.

Example: Deployment of IoT-Based Cold Storage Monitoring in Uganda.

MODULE 3

CHOOSING A RESEARCH TOPIC

Decision Tree

Do you have original results?



Yes ▼ No

Research Article



Are you comparing existing studies?



Yes ▼

Survey Paper



Summarizing literature?



Yes ▼

Review Paper



Good Research Topics

AI

Explainable AI for Healthcare
AI-Based Crop Disease Detection
Edge AI for IoT Devices

Networking

LoRa Coverage Optimization
Adaptive Network Selection
5G Rural Connectivity

IoT

Smart Beehive Monitoring
Cold Storage Monitoring
Smart Irrigation Systems

Characteristics

- ✓ Novel
- ✓ Relevant
- ✓ Feasible
- ✓ Impactful
- ✓ Publishable

MODULE 4

FINDING AND READING LITERATURE

Why Literature Matters

Literature Review helps researchers:

- Avoid duplication
- Understand existing solutions
- Identify gaps
- Improve methodology
- Justify research
- Without Literature Review
- You may solve a problem already solved five years ago.

Finding Papers

- Google Scholar
- IEEE Xplore
- Springer
- Elsevier
- MDPI
- Using Search Strings

Examples

Reading Papers Efficiently

- Abstract
- Figures
- Results
- Conclusion

Literature Review Matrix

Author	Year	Technology	Method	Findings	Gap
Smith	2022	WiFi	Simulation	High throughput	Limited range
Ali	2023	LoRa	Experiment	Long range	Low data rate
Namugenyi	2024	GSM	Field Study	Good coverage	High cost

Identifying Research Gaps

Research Gap = Missing Knowledge

Example

Researchers studied:

WiFi

GSM

LoRa

But no one compared all three technologies under identical conditions.

Gap: Comparative evaluation of WiFi, GSM and LoRa for beehive monitoring.

Gap Analysis Example

Gap: No integrated solution exists.

Study	WiFi	GSM	LoRa	AI
Study A	✓	X	X	X
Study B	X	✓	X	X
Study C	X	X	✓	X
Proposed Study	✓	✓	✓	✓

MODULE 5

JOURNAL SELECTION

Common Literature Review Mistakes

✗ Listing papers one by one

✗ No comparison

✗ No research gap

✗ Old references only

✗ Weak critical analysis

Better Practice: Compare studies using tables and matrices.

Why journal selection matters?

Wrong journal = Immediate rejection.

Consequences

- Long delays
- Wasted APCs
- Multiple resubmissions

Goal: Find a journal whose scope perfectly matches your research.

Understanding Journal Scope

Scope describes:

Topics accepted

Methods accepted

Audience

Example

Topic:

LoRa Communication Networks

Suitable:

Networking journals

IoT journals

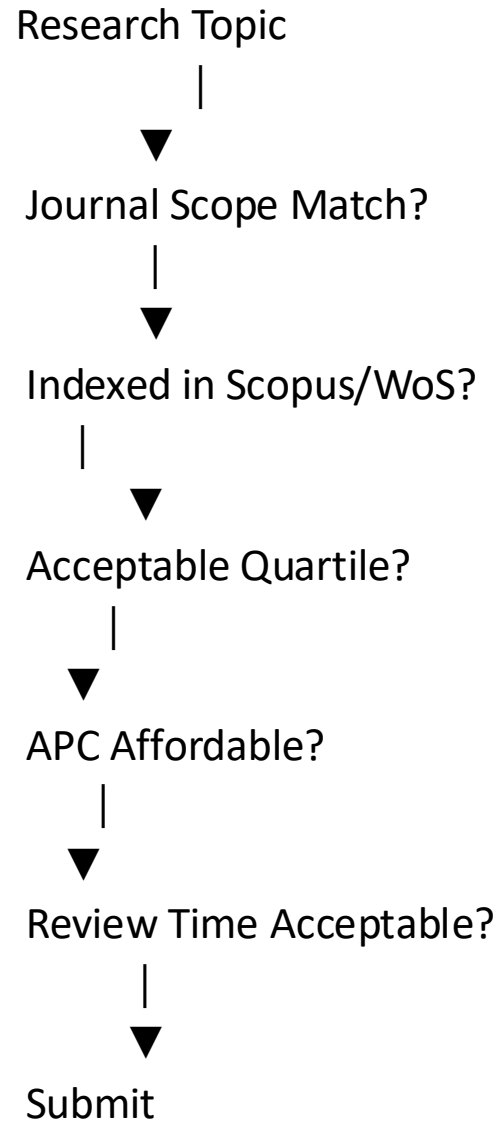
Unsuitable:

Medical Imaging journals

Matching Topic to Journal

Topic	Suitable Journal Category
AI	Artificial Intelligence
Networking	Communication Networks
IoT	Internet of Things
Cybersecurity	Information Security
Data Science	Analytics & Data Science

Journal Selection Flow Chart



Understanding Indexing

Scopus: Large citation database.

Web of Science: Highly selective indexing database.

Google Scholar: Broad scholarly search engine.

DOAJ: Directory of Open Access Journals.

Importance

Indexed journals are more visible and respected.

Journal Metrics

Impact Factor: Average citations per article.

CiteScore: Scopus citation metric.

SJR: Measures journal prestige.

SNIP: Field-normalized impact.

H-index: Productivity and citation influence.

Understanding Quartiles

Quartiles rank journals within a field.

Quartile	Meaning
Q1	Top 25%
Q2	25–50%
Q3	50–75%
Q4	Bottom 25%

Recommendation for Students

Level	Target
Undergraduate	Q2–Q4
MSc	Q2–Q3
PhD	Q1–Q2
Academic Staff	Q1

Journal Quality Assessment Rubric

Criterion	Marks
Scope Match	25
Indexing	20
Quartile	15
Impact Metrics	10
Publisher Reputation	10
APC Affordability	10
Review Time	5
Acceptance Probability	5
Total	100

MODULE 6

JOURNAL STRUCTURE

Typical Structure of a Journal Paper

Why Structure Matters? Editors and reviewers expect a standard format.

Typical Structure

Title



Authors & Affiliations



Abstract



Keywords



Introduction



Literature Review



Methodology



Results



Discussion



Conclusion



References

Key Message: A good paper tells a story:

Problem → Solution → Evidence → Contribution

The Title Section

Purpose: The title is the first thing readers see.

Characteristics

- ✓ Specific
- ✓ Searchable
- ✓ Concise
- ✓ Descriptive

Poor Example: "Network Study"

Better Example: "Performance Evaluation of LoRa, WiFi and GSM Networks for Smart Beehive Monitoring"

Author Information

Components

Author names

Institution

Department

Email

ORCID (optional)

Example

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Makerere University

Kampala, Uganda

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Abstract

Purpose: Summarizes entire paper.

Structure

Problem

Method

Results

Contribution

Good Example: "This study evaluates LoRa, GSM and WiFi communication technologies for smart beehive monitoring. Experimental results indicate that LoRa provides superior range while WiFi achieves the highest throughput. An adaptive communication architecture is proposed."

Poor Example: "This paper discusses networks and gives results."

Keywords

Purpose: Improve discoverability.

Guidelines

5–8 keywords.

Example

IoT

LoRa

GSM

WiFi

Smart Agriculture

Wireless Networks

Machine Learning

Introduction

Purpose: Explain why the study matters.

Components

Background

Problem Statement

Research Gap

Objectives

Contributions

Example: The increasing adoption of IoT in agriculture has created demand for reliable communication systems in rural environments.

Literature Review

Purpose: Show understanding of existing work.

Questions

What has been done?

What has not been done?

Why is your work needed?

Literature Review Matrix

Author	Technology	Method	Findings	Gap
Smith	WiFi	Simulation	High throughput	Short range
Ali	GSM	Field Study	Wide coverage	Expensive
Chen	LoRa	Experimental	Long range	Low data rate

Gap: No comparative study evaluating all three technologies under identical conditions.

Purpose: Explain how research was conducted.

Methodology Flow

Problem Definition



System Design



Architecture Development



Data Collection



Experiments



Analysis



Results

Methodology Overview

Architecture Diagram Example

IoT Communication Architecture

Sensors



ESP32



LoRa/WiFi



Gateway



Cloud Server



Mobile Application

Discussion: Explain each layer and its role.

Flowchart Example

System Operation

Start



Collect Data



Process Data



Store Data



Display Results



End

Purpose: Demonstrates system logic.

Data Collection

Types

Measurements

Experiments

Surveys

Datasets

Examples

Networking:

Throughput

RSSI

Packet Loss

Latency

AI:

Accuracy

Precision

Recall

F1 Score

Results Section

Purpose: Present findings only.

Include

Tables

Figures

Graphs

Statistics

Avoid

Interpretation

That belongs in Discussion.

Example Results Table

Distance (m)	Throughput (Mbps)
10	25
20	22
30	19
40	16
50	12

Engineering Graphs

Common Graph Types

Graph	Use
Line	Performance trends
Bar	Comparisons
Scatter	Correlations
Histogram	Distributions
Boxplot	Variability
Radar	Multi-metric comparison
Heatmap	Coverage analysis

Publication Quality Figures

Requirements

- ✓ High resolution
- ✓ Clear labels
- ✓ Readable fonts
- ✓ Proper captions
- ✓ Referenced in text

Example

Figure 4.1: Throughput Performance versus Distance

Graph Interpretation

Weak Interpretation:

"Figure 5 shows throughput."

Strong Interpretation:

"Figure 5 shows throughput decreasing with distance due to signal attenuation and reduced signal quality."

Equations and Mathematical Models

Why Use Equations?

Precision

Reproducibility

Scientific rigor

Example

Throughput = Data Received / Time.....(1)

Engineering Equation Examples

Throughput

Packet Loss

Latency

Latency=Receive Time–Send Time

RSSI

Measured in dBm.

Discussion Section

Purpose: Interpret results.

Questions

Why did it happen?

What does it mean?

How does it compare with previous studies?

Example: LoRa achieved greater coverage than WiFi due to lower operating frequencies and improved propagation characteristics.

Conclusion Section

Purpose: Summarize study.

Include

Findings

Contributions

Limitations

Future work

Avoid

Introducing new results.

Reference Section (IEEE Style)

Journal Article

[1] E. E. Namugenyi, "Adaptive Communication Architecture for Smart Beehive Monitoring," Journal Name, vol. 5, no. 2, pp. 10–20, 2025.

Conference Paper

[2] E. E. Namugenyi, "Performance Evaluation of LoRa Networks," Proc. IEEE Conference, pp. 45–50, 2025.

Book

[3] A. Tanenbaum, Computer Networks, 5th ed. Pearson, 2011.

Website

[4] IEEE Standards Association. Available: <https://standards.ieee.org>

MODULE 7

MANUSCRIPT FORMATTING

IEEE Manuscript Formatting Guide

Standard Format

Two-column layout

Times New Roman

10 pt font

Numbered references

Figure captions below figures

Table captions above tables

Numbered equations

Figures, Tables and Captions

Figures

Figure 3.1: IoT Architecture

Tables

Table 3.1: Throughput Measurements

Rules

Always cite them in text.

MODULE 8

PEER REVIEW AND PUBLICATION

Submission and Review Process

Author



Editor Screening



Reviewers



Decision



Revision



Acceptance



Publication

Reviewer Comments and Responses

Reviewer

"The novelty of the architecture is unclear."

Response

"We thank the reviewer. Section 1 has been revised to clarify the novelty and contributions."

Common Reasons for Rejection

- ✗ Poor English
- ✗ Weak novelty
- ✗ Poor methodology
- ✗ Insufficient experiments
- ✗ Poor figures
- ✗ Incorrect journal scope
- ✗ Plagiarism

MODULE 9

PUBLICATION ETHICS

Plagiarism

Definition: Using another person's work without proper attribution.

Types

Copy-paste

Mosaic plagiarism

Self-plagiarism

Prevention

Cite sources

Paraphrase properly

Use reference managers

Authorship

Who Qualifies?

Someone who:

- ✓ Contributed substantially
- ✓ Participated in writing
- ✓ Approved final manuscript

Problems

Gift authorship

Ghost authorship

Data Fabrication and Falsification

Fabrication: Making up data.

Falsification: Manipulating results.

Consequences

Retractions

Reputation damage

Academic sanctions

MODULE 10

PUBLICATION READINESS

Publication Readiness Rubric

Component	Marks
Title	5
Abstract	10
Introduction	10
Literature Review	10
Methodology	15
Results	15
Discussion	10
Figures & Tables	10
References	10
Formatting	5
Total	100

Final Publication Checklist and Student Roadmap

Checklist

- ☐ Title optimized
- ☐ Abstract complete
- ☐ References verified
- ☐ Figures numbered
- ☐ Tables numbered
- ☐ Equations numbered
- ☐ Ethics checked
- ☐ Journal selected
- ☐ Scope verified

Student Roadmap

Internship



Final Year Project



Conference Paper



Journal Paper



MSc Research



PhD Research



Academic / Research Career



Questions and Discussion



Thank You